Mario A. Esquivel III

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Education

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science in Computer Science — *Aug. 2023 – May 2027*
Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Probability, Discrete Math, Machine Learning

Experience

IT Backend Developer — Jan. 2025 – Present
Georgia Institute of Technology | Part-time — *Atlanta, GA*

- Improved backend support efficiency by automating account provisioning, system access, and device setup workflows across Windows, macOS, and Linux environments with Python scripts, supporting 50+ users weekly.
- Reduced manual troubleshooting time by 30% by building scripting-based solutions for recurring operational tasks and using Vim to quickly modify system files, configurations, and backend support tools.
- Increased deployment consistency by standardizing workstation configurations and validating network-connected systems, contributing to 100+ successful device setups weekly.

Projects

YouTubeTrades | Next.js, React, TypeScript, Tailwind CSS, YouTube API, Finnhub — June 2026

- Built a stock-hype dashboard that ranks equities by aggregate audience exposure by extracting stock mentions from finance YouTube titles, descriptions, and transcripts using a rules-based system covering approximately 115 known tickers and company names.
- Kept market-data requests within Finnhub's 60-request-per-minute API limit by implementing 10-minute quote caching, 30-minute channel caching, disk-cached transcripts, and bounded concurrency of approximately four channels at a time.

Brain Tumor Neural Network Classifier | Python, PyTorch, torchvision, ResNet18 — June 2026

- Achieved 94.7% classification accuracy on 1,600 held-out brain MRI scans by fine-tuning a pretrained ResNet18 across a 7,200-image dataset containing glioma, meningioma, pituitary tumor, and no-tumor classes.
- Improved accuracy under simulated sensor noise from 51.2% to 93.9% and quarter-resolution images from 62.8% to 91.9% by engineering robust crop, brightness, contrast, blur, and noise augmentation during training.
- Reached 100% recall for no-tumor scans and 99.0% recall for pituitary tumors by building an evaluation pipeline with early stopping, best-model checkpointing, confusion matrices, per-class precision/recall, and confidence-based inference.

Noted | Next.js, React, TypeScript, Supabase, OpenAI, Playwright — June 2026

- Reduced manual calendar-planning effort by an estimated 70% by building an AI-powered student planner that imports Canvas coursework and generates personalized study schedules using natural-language, image, and voice input.
- Protected student and AI-generated data across all API endpoints by implementing Supabase authentication and Row-Level Security, Zod input validation, security headers, and rate limits of 60 requests/minute for standard routes and 10 requests/minute for AI-intensive routes.

Leadership, Club, & Volunteer Work

Coding Instructor — Jan. 2026 – Present
Black Student Computing Organization @ Georgia Tech — *Atlanta, GA*

- Taught 20+ students at ABC Learning Academy introductory programming concepts using Scratch, helping them build their first interactive games and animations.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C, C++, C#, Go, Rust, Kotlin, SQL, Bash/Shell, R, Swift, Ruby, PHP, Scala, HTML/CSS, Assembly
Frameworks/Libraries: React.js, Next.js, Node.js, Express.js, Django, Spring Boot, NumPy, JAX, Pandas, OpenCV, PyTorch, TensorFlow
Databases: PostgreSQL, MySQL, MongoDB, Firebase, Supabase
Tools/Platforms: Git, GitHub, Docker, AWS, REST APIs, Vim, Playwright, OpenGL, Unity, Unreal Engine, Jira, Power BI